



University of Information Technology and Sciences

CodeLearn

Your Personal Programming Mentor

Supervised By



Mr. Al-Imtiaz

Associate Professor & Head Department of CSE, UITS

Submitted By



Muhammad Imtiaz Ahmed

ID – 0432410005101138

Foyze Ahmed

ID – 1944651070



Project Objectives

- To effectively connect programming students with **qualified tech teachers**.
- To streamline session **booking, payment, and management**.
- To provide an **integrated, real-time chat** system for users.
- To create a **secure, localized** platform for the **Bangladeshi market**.
- To deliver a comprehensive **admin panel** for platform oversight.

Related Works

1. Ten Minute School (10MS)

- **Comprehensive**
- **pre-recorded video courses and live classes** covering academic curriculum (JSC, SSC, HSC, university admission), skill development, and professional training.

2. Shikkhok.com

- Platforms like Shikkhok.com or major local tutoring services that include an online component.

3. Local Freelancing Platforms (e.g., as used by Bangladeshi developers)

- While not dedicated **tutoring sites**
- **Online groups**, occasionally offering coaching or mentorship services.

Feasibility: Market Analysis

<CodeLearn>



200,000+

Programming Students
in Bangladesh



5,000+

Potential Tech
Instructors



₹1,000 Crore

EdTech Market (2025
Est.)

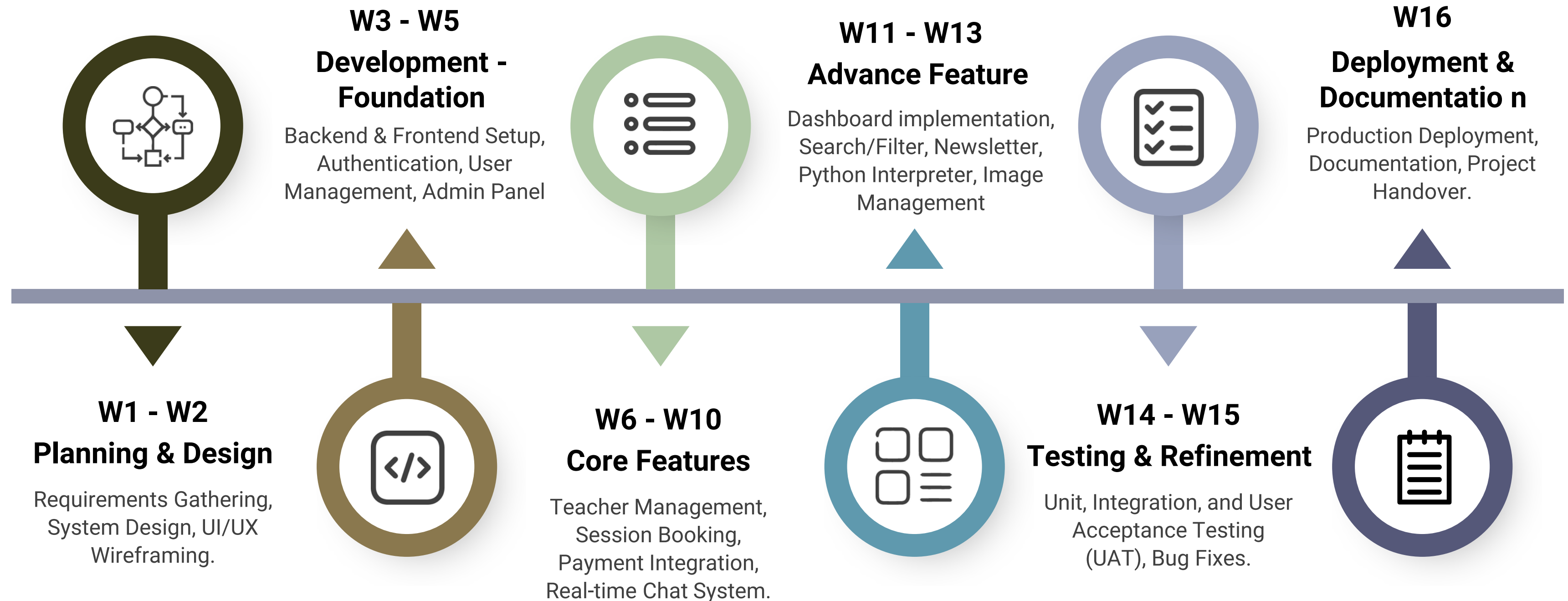


₹50–100 Cr

Target Addressable
Market

Methodology (16-Week Plan)

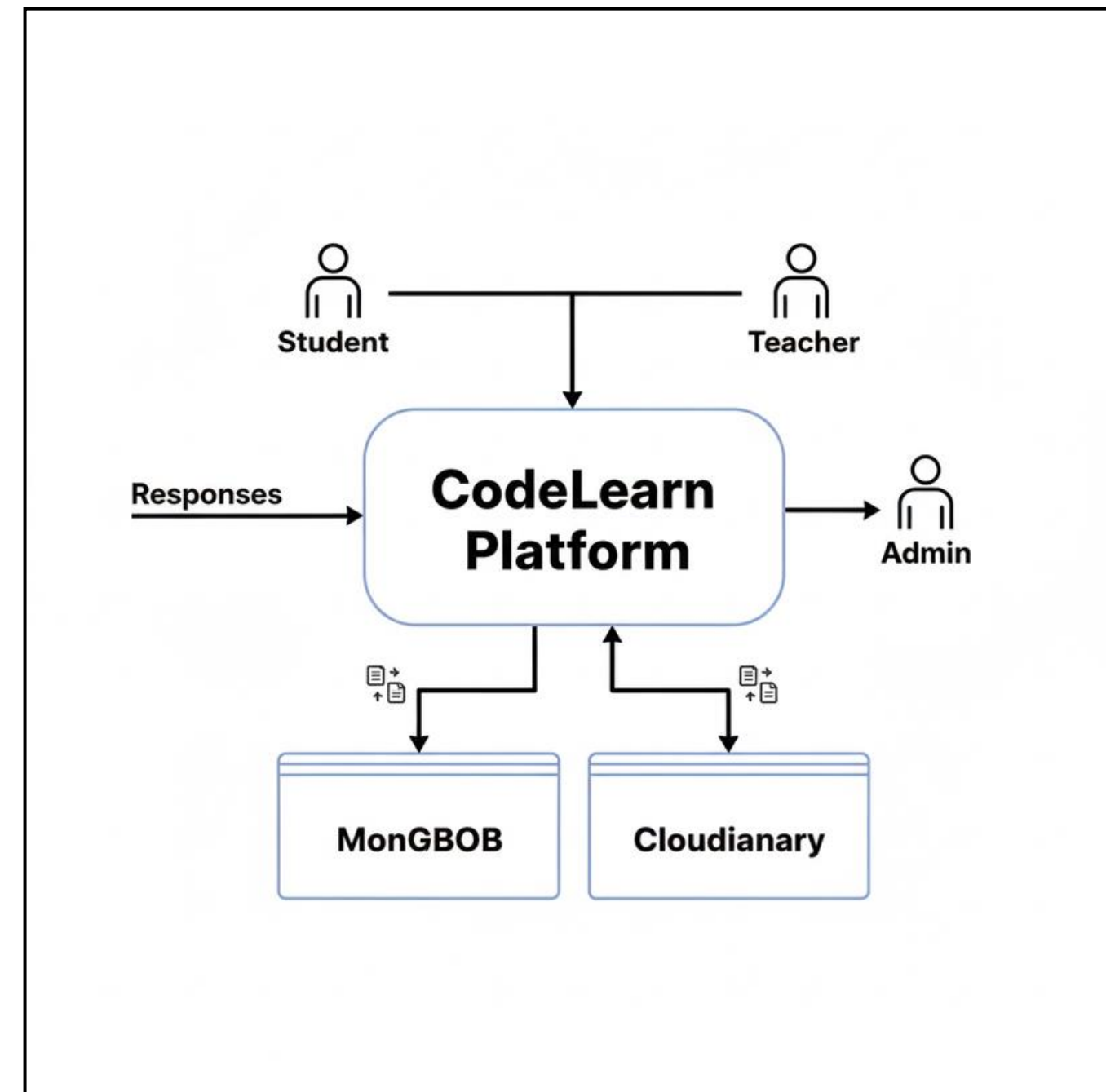
<CodeLearn>



Data Flow Diagram (DFD)

<CodeLearn>

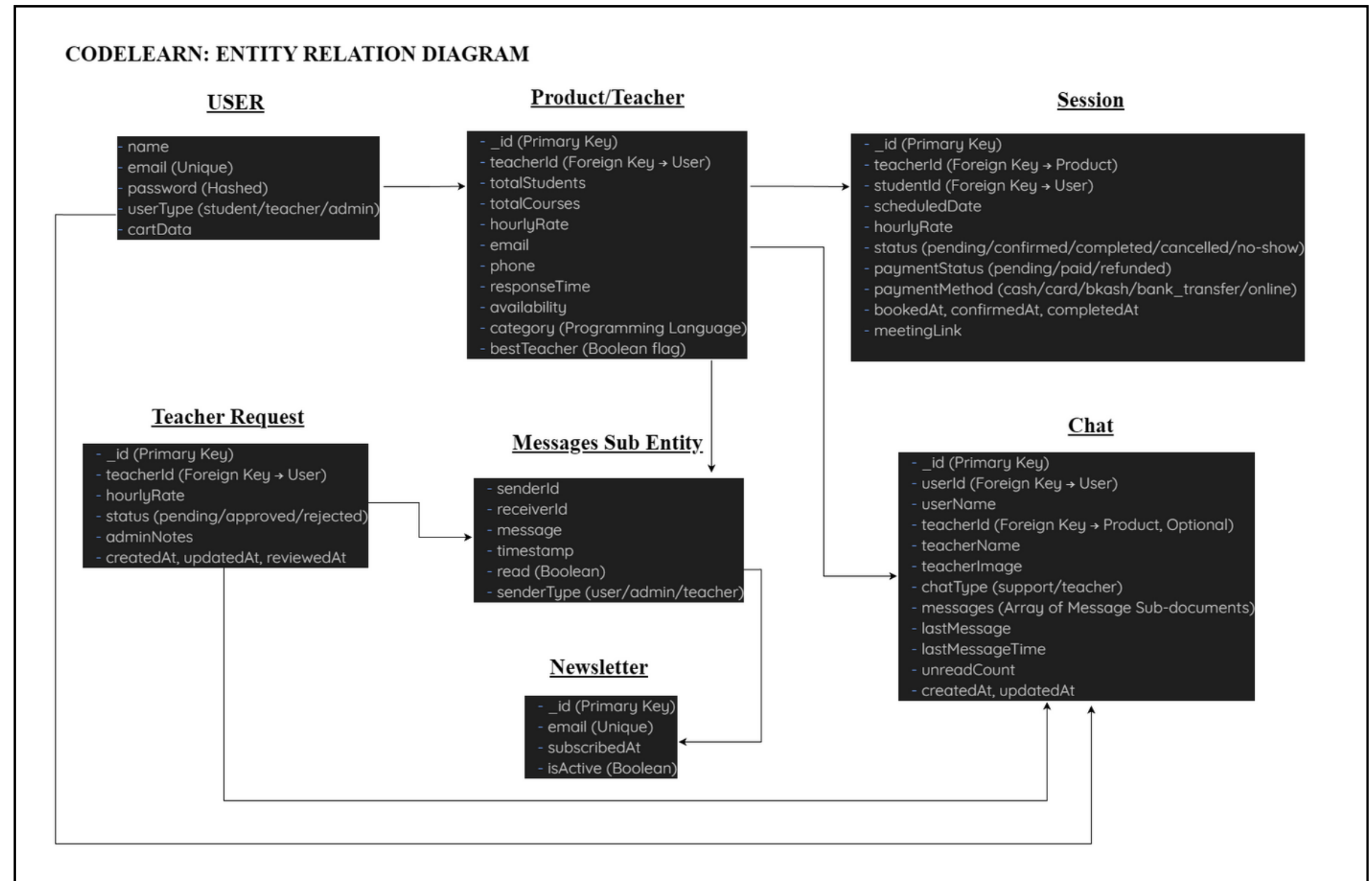
- **System/Process:** CodeLearn Platform
- **External Entities/Users:** Student, Teacher, Admin
- **Data Stores:** MonGBOB, Cloudianary
- **Data Flow:** Responses



Entity Relation Diagram

<CodeLearn>

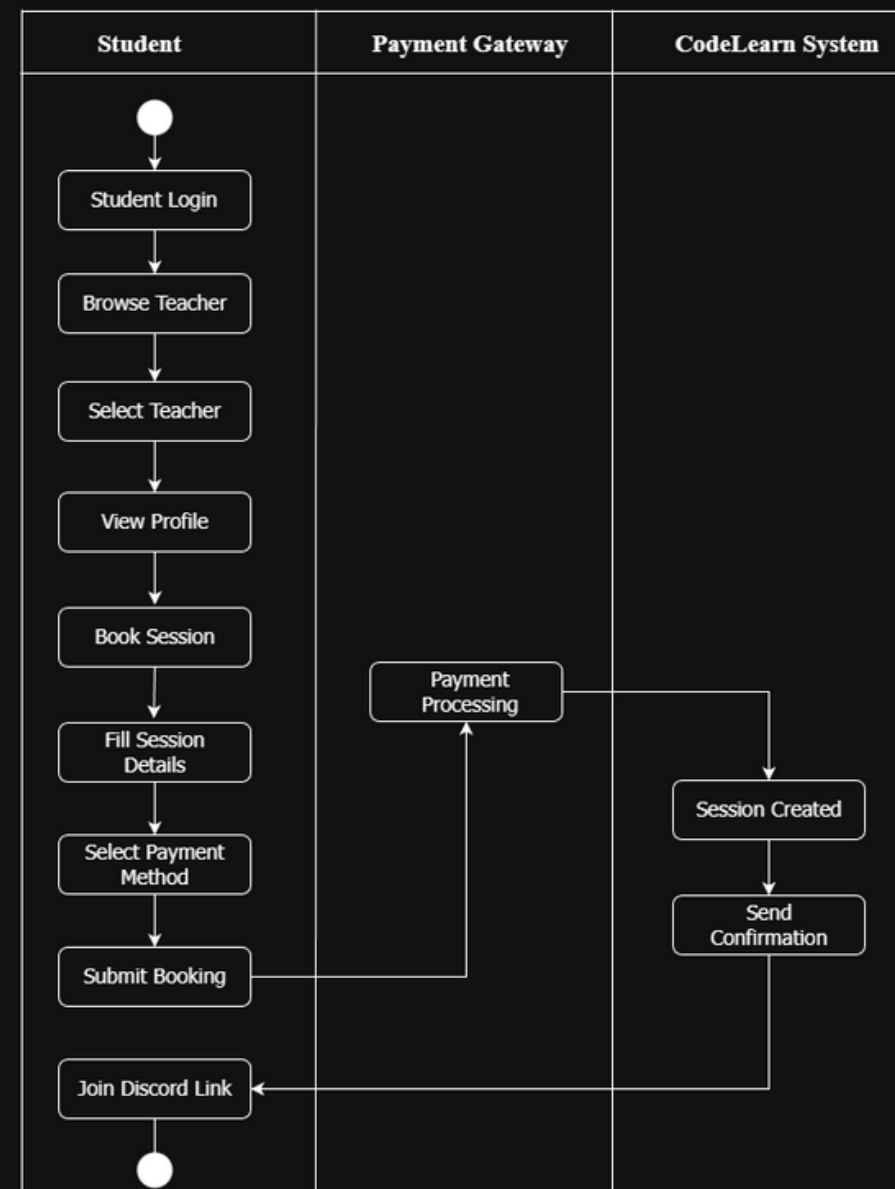
- **User:** The base model for all roles.
- **Product/Teacher:** Teacher-specific details, linked to User.
- **Session:** Records of booked events, linking a Teacher and Student.
- **Chat:** Manages communication channels.
- **Teacher Request:** Handles the vetting process for new teachers.



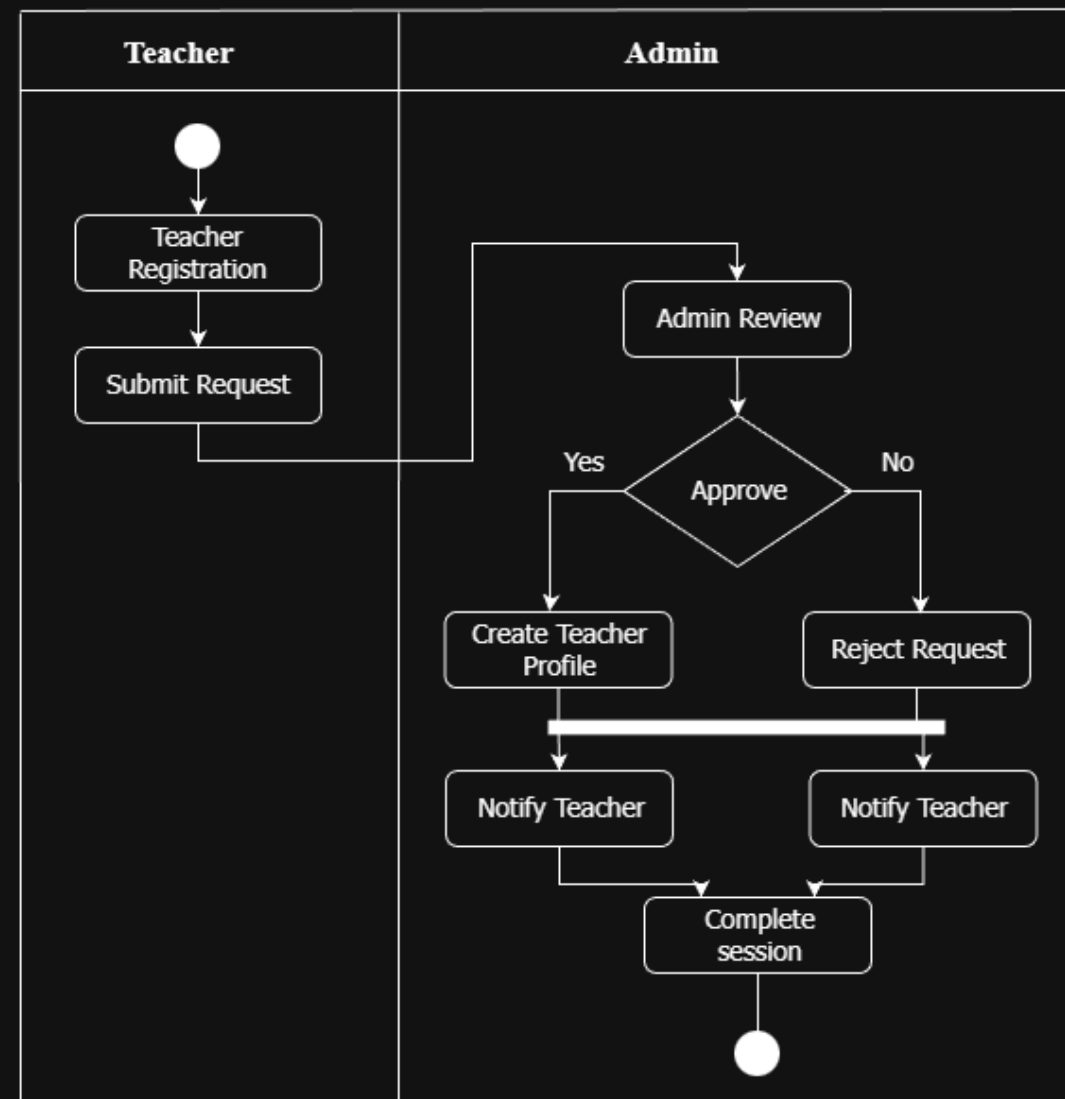
Activity Diagram

<CodeLearn>

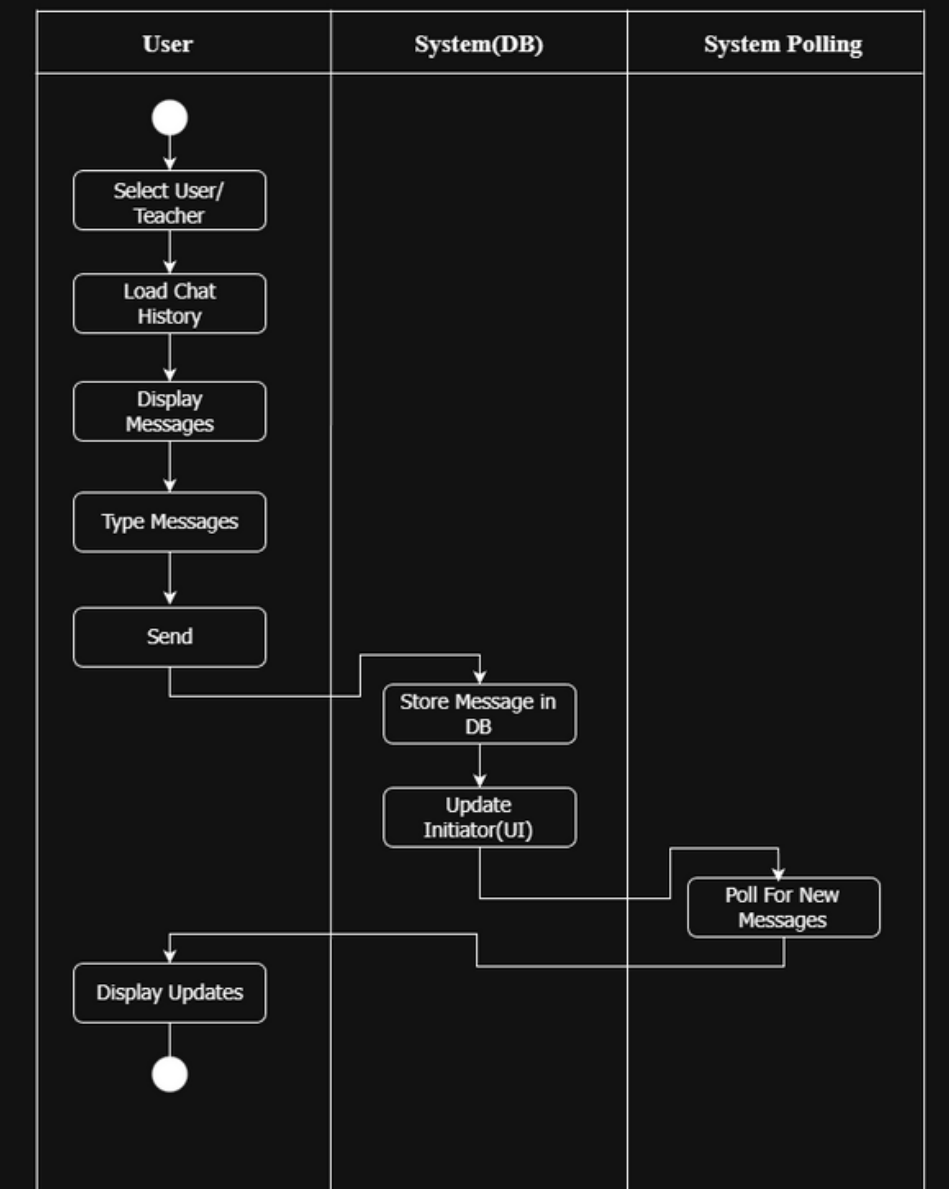
CODELEARN : USER ACTIVITY DIAGRAM



CODELEARN : TEACHER ACTIVITY DIAGRAM



CODELEARN : CHAT ACTIVITY DIAGRAM



User Section (Student)

<CodeLearn>

The screenshot shows a web browser window with the URL `localhost:5173/login`. The browser's address bar and tabs are visible at the top. The page itself has a dark header with the CodeLearn logo and navigation links: Home, Teachers, About Us, and Contact. Below the header, there are two tabs: 'Student' (selected) and 'Teacher'. The main content area is titled 'Login —' and contains two input fields: 'email@gmail.com' and 'Password'. Below these fields are links for 'Forgot Password' and 'Create Account'. A large black 'Sign In' button is positioned below the links. The footer section is divided into three columns: 'CodeLearn' (with a description of the platform), 'Quick Links' (with links to Home, About Us, Contact, and Teachers), and 'Get in Touch' (with contact information: 01331759272, support@codelearn.com, and Live Chat Support). The bottom of the screen shows a Windows taskbar with various application icons and a system tray displaying the date and time (12:13 PM, 11/8/2025).

CodeLearn

Home Teachers About Us Contact

Student Teacher

Login —

email@gmail.com

Password

[Forgot Password](#) [Create Account](#)

Sign In

<> CodeLearn

CodeLearn is your premier online platform for connecting with expert programming teachers and tech mentors. We connect learners with highly qualified tech educators to accelerate your technology learning journey.

Quick Links

- Home
- About Us
- Contact
- Teachers

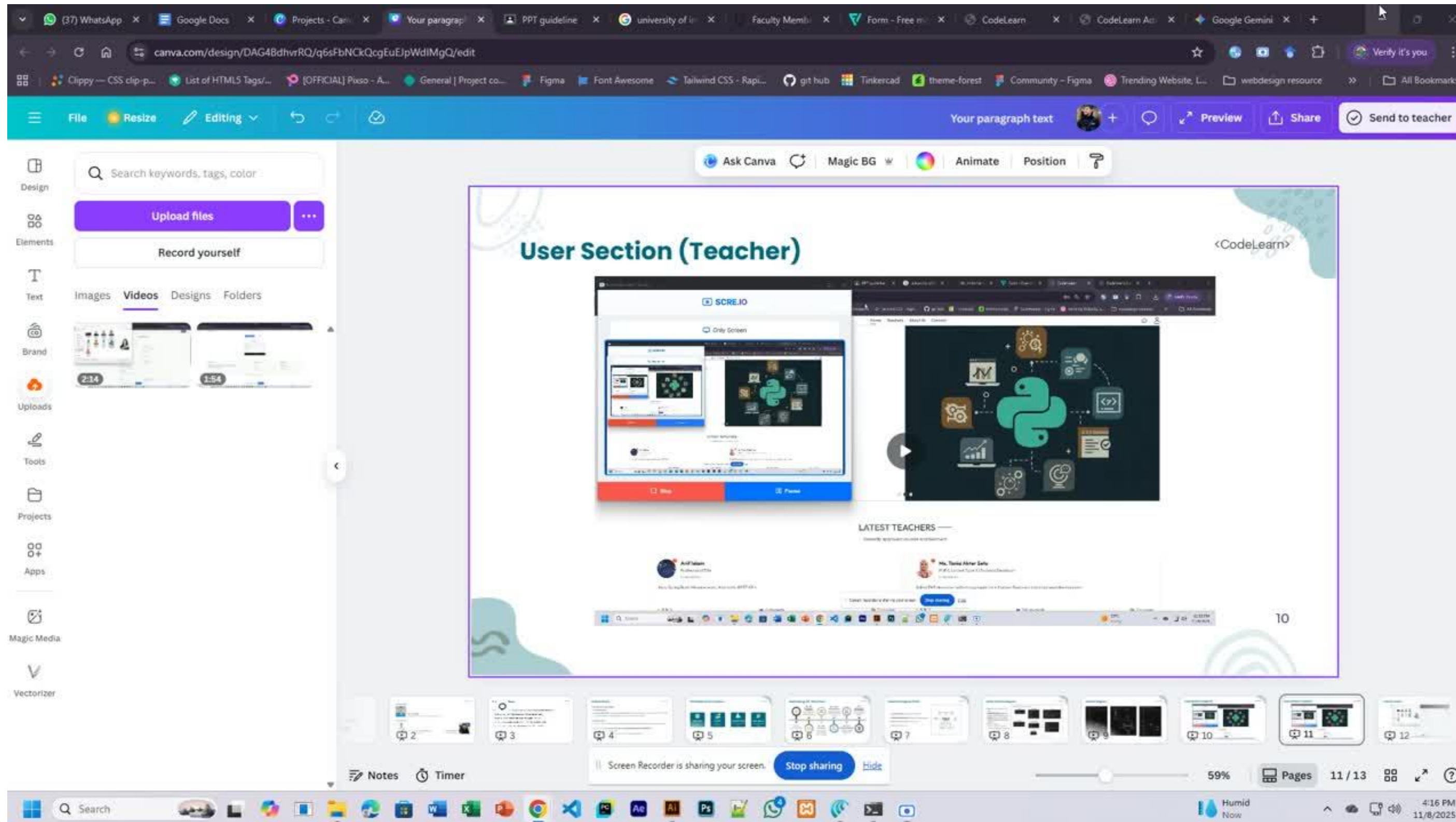
Get in Touch

- 01331759272
- support@codelearn.com
- Live Chat Support

Copyright 2025© codelearn.com - All Right Reserved

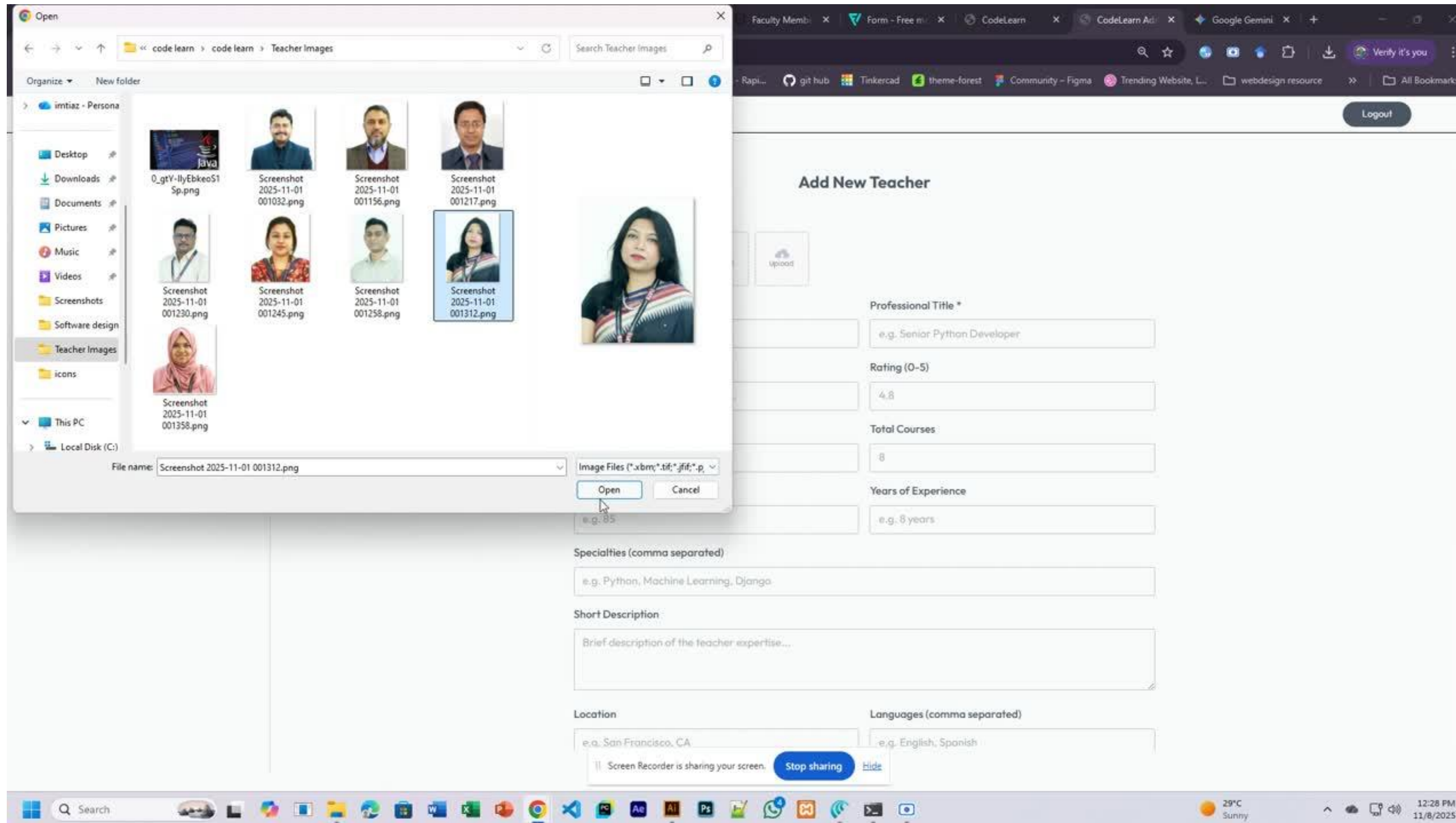
User Section (Teacher)

<CodeLearn>



Admin Section

<CodeLearn>



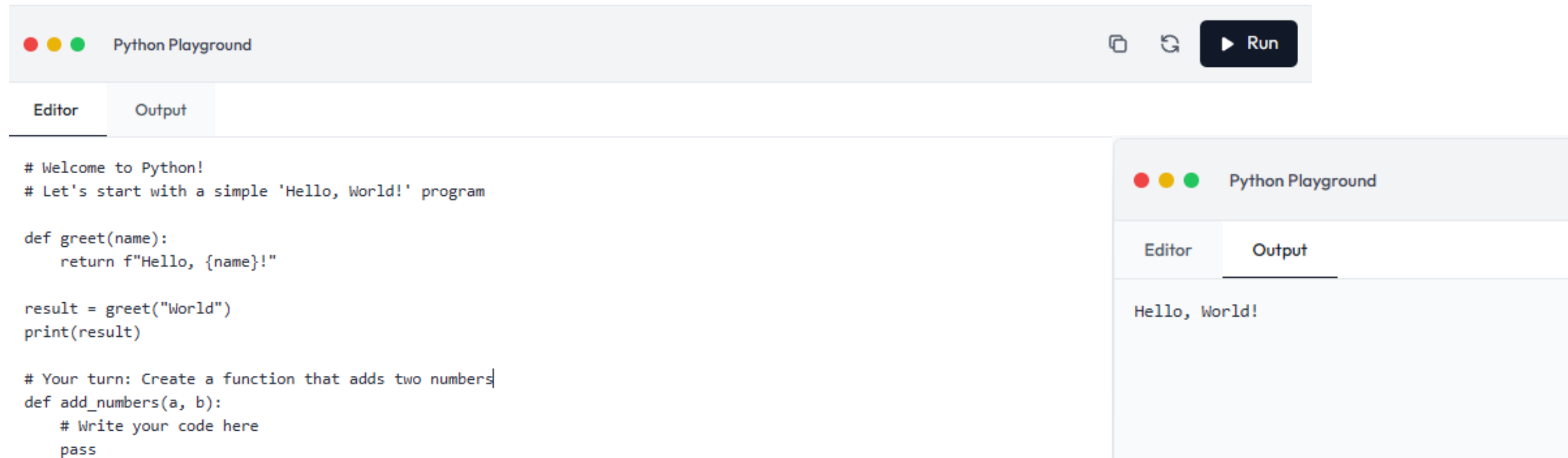
Admin Section

<CodeLearn>

The screenshot displays the CodeLearn Admin interface with a 'Screencast.io' recording overlay. The overlay, titled 'What do you want to capture?', offers three options: 'Only Screen', 'Screen + Camera' (marked 'Coming Soon'), and 'Only Camera'. Below this, the 'Record audio?' section provides four choices: 'None', 'Microphone', 'System + Mic.', and 'System'. A large blue 'Start Recording' button is at the bottom of the overlay. The background shows the 'Add New Teacher' form, which includes fields for 'Image URL', 'Professional Title *', 'Rating (0-5)', 'Total Courses', 'Years of Experience', 'Specialties (comma separated)', 'Short Description', 'Location', and 'Languages (comma separated)'. A 'Logout' button is visible in the top right corner of the admin panel. The Windows taskbar at the bottom shows the time as 4:36 PM on 11/8/2025.

Python Playground

<CodeLearn>



The image shows a web-based Python Playground interface. It consists of a main editor window and a smaller output window. The main editor window has a title bar with three colored circles (red, yellow, green) and the text "Python Playground". Below the title bar are two tabs: "Editor" and "Output". The "Editor" tab is active, showing the following Python code:

```
# Welcome to Python!  
# Let's start with a simple 'Hello, World!' program  
  
def greet(name):  
    return f"Hello, {name}!"  
  
result = greet("World")  
print(result)  
  
# Your turn: Create a function that adds two numbers  
def add_numbers(a, b):  
    # Write your code here  
    pass
```

The "Output" tab is also visible, showing the result of the first function call:

```
Hello, World!
```

The smaller output window on the right has a similar title bar and tabs. Its "Output" tab is active, displaying the same "Hello, World!" message.

Write, run, and debug Python code right in your browser. No setup required!

Socio-economic Impact

<CodeLearn>

Market Analysis

Factor	Details
Programming Students	200,000+ in Bangladesh
Tech Teachers	5,000+ potential instructors
Institutions (B2B)	Coding academies, universities
EdTech Market (BD)	~৳1,000 crore (2025 est.)
Tech Training Segment	~৳150 crore
Target Addressable Market	৳50-100 crore

Revenue Streams

Model	Description	Example / Rate
Commission-Based	CodeLearn takes 15-20% from each completed session	If teacher charges ৳5,000 → CodeLearn earns ৳750-৳1,000
Subscription Plans	Monthly subscriptions for teachers & students	Teacher Premium: ৳500/month; Student Premium: ৳300/month
Featured Listings	Premium placement for teachers	৳2,000/month per teacher in "Top Teachers" section

Limitations

<CodeLearn>



Web application only (no native mobile app).



Strictly one-on-one sessions (no group classes).



No automated AI/ML mentor matching.

Future Work

<CodeLearn>



Develop native mobile apps for iOS and Android.



Introduce group sessions and webinar feature.



Implement AI for personalized mentor matching.



Integrate a wider range of international and local payment gateways.



Expand B2B (Business-to-Business) model for educational institutions.



Add live video session and interactive whiteboard features.

Conclusion

<CodeLearn>

CodeLearn fulfills its objective of connecting **learners and mentors**. The project showcases strong engineering competencies, with **86.6% PO attainment**. Financial projections highlight strong commercial **viability and sustainability**.